

Charlotte Chen

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Research Interests

Computer Architecture: Domain-specific accelerators, power-performance characterization, hardware-software co-design
SoC Design: Heterogeneous SoC integration, design-space exploration, RTL-to-silicon implementation

Education

Columbia University

New York, NY

B.S. in Computer Engineering | Major GPA: 3.86/4.0

Sept. 2024 – May 2026

Relevant Coursework: VLSI Design Lab (Tapeout), Computer Architecture, SoC Platforms, Advanced Logic Design

Honors: Dean's List, Tau Beta Pi Honor Society

Grinnell College

Grinnell, IA

B.A. in Computer Science, B.A. in Japanese | GPA: 3.96/4.0 (Major: 4.0/4.0)

Aug. 2021 – May 2024

Honors: Dean's List

Research Experience

Power and Performance Analysis of SoC Architectures

Jan. 2025 – Present

Undergraduate Researcher (Advisor: Prof. Simha Sethumadhavan, Mentor: Annie Peng) | Columbia University New York, NY

- Characterized NVIDIA Jetson Nano SoC and GPU power-performance behavior via hardware performance counters, establishing energy-delay product (EDP) baselines across heterogeneous compute units under diverse workload conditions.
- Designed and executed targeted microbenchmarks varying execution unit utilization and memory access patterns to empirically quantify DVFS sensitivity curves and power-gating efficiency tradeoffs from measured hardware data.

GreenerSKU: Lower Carbon Datacenter Operations Research

May 2024 – Present

Undergraduate Research Assistant (PI: Prof. Yang Katie Zhao) | Univ. of Minnesota Twin Cities Minneapolis, MN

- Built and calibrated a datacenter carbon emission simulation framework to systematically measure the joint effects of dynamic workload scheduling and cooling strategies on total carbon footprint across heterogeneous server configurations.
- Refactored the Python simulation engine to eliminate concurrency bottlenecks, achieving high-fidelity alignment between simulated and empirical datacenter power traces and cross-validating model predictions against real-world deployment topologies.

Autonomous Driving Algorithms Research

Dec. 2023 – April 2024

Undergraduate Research Assistant (PI: Prof. Nuria González-Prelcic) | NC State University Raleigh, NC

- Implemented hardware-in-the-loop path-planning systems and evaluated dynamic vehicle behavior through structured test scenarios, quantifying tradeoffs between computational safety margins and routing efficiency.
- Measured collision time-to-contact (TTC) margins and algorithmic latency across a comprehensive suite of real-world driving scenarios, producing safety-critical performance metrics to validate algorithm correctness.

Publications & Presentations

C. Chen, "Power and Performance Analysis of GPU under DVFS." *University Research Presentation, Columbia University*, Spring 2025. [\[View Slides\]](#)

Selected Hardware & Architecture Projects

TSMC 65nm Tape-Out: N-body Simulation Accelerator | *Verilog, Innovus, PrimeTime* Jan. 2026 – Present

- Architected a $1mm^2$ reconfigurable systolic array with four parallel Processing Elements (PEs) to offload $O(N^2)$ pairwise particle interactions from the host CPU, targeting 100MHz operation at $< 50mW$ power consumption.
- Implemented a programmable Look-Up Table (LUT) force evaluation engine, updatable dynamically via SPI, to support non-linear physics kernels including gravitational and Lennard-Jones potentials.
- Executed full RTL-to-GDSII physical design flow (Design Compiler synthesis, Innovus Place & Route) with DFT scan-chain insertion; validated Q4.12 fixed-point arithmetic precision against a Python golden model through systematic co-simulation, quantifying rounding error across the force computation pipeline.

Superscalar Out-of-Order RISC-V Processor | *SystemVerilog, VCS, Verdi* Jan. 2025 – Present

- Designed a 32-bit superscalar out-of-order RV32IM processor in SystemVerilog based on the P6/R10K microarchitecture, supporting speculative execution, early tag broadcast, and Simultaneous Multithreading (SMT).
- Configured aggressive microarchitecture with 64-entry register renaming (FreeList/MapTable), 16-entry Reservation Station (RS), and 32-entry Reorder Buffer (ROB) to maximize instruction-level parallelism.
- Integrated parallel functional units (3 ALUs, 2 Loads, 2 Multipliers, 1 Branch) and a hybrid branch predictor (BTB + Bimodal) to improve IPC; validated functional correctness by automated differential testing against a C++ golden model across exhaustive instruction sequences.

TPU-like Accelerator Integration in ESP | *SystemVerilog, SoC Design* Oct. 2024 – Dec. 2024

- Conducted systematic design-space exploration across tiling factors and dataflow configurations for CNN convolutional layer accelerators, measuring effective latency against FPGA LUT/DSP resource constraints to identify Pareto-optimal operating points.
- Integrated the accelerator IP into the Embedded Scalable Platforms (ESP) SoC via AXI interfaces with DMA burst transfers; profiled end-to-end memory bandwidth utilization to maximize throughput.
- Benchmarked system performance under varied hardware configurations and deployed the final SoC to FPGA for application-level hardware validation, comparing measured throughput and power against simulation estimates.

Computer Architecture-Level Large Data Optimization | *C/C++, Python, gem5* Fall 2024

- Analyzed performance bottlenecks at the memory hierarchy and instruction pipeline levels in large-scale data processing workloads, using architectural simulation to isolate the contribution of cache capacity, bandwidth, and instruction-level parallelism.
- Profiled and compared microarchitectural configurations to quantify tradeoffs between throughput and latency, producing a data-driven evaluation of optimization strategies for data-intensive workloads.

Teaching Experience

Computer Architecture Teaching Assistant Sept. 2025 – Dec. 2025

Department of Computer Science, Columbia University

New York, NY

- Guided students through hardware design labs, assisting with advanced Verilog debugging, timing analysis, and out-of-order processor simulation methodologies.
- Facilitated discussion sections to reinforce complex microarchitecture concepts including pipelining, caching policies, and hazard resolution.

Computer Architecture Teaching Assistant Jan. 2024 – May 2024

Department of Computer Science, Grinnell College

Grinnell, IA

- Led multiple weekly lab and discussion sessions to reinforce computer architecture fundamentals, assembly programming, and single-cycle datapath execution.

Technical Skills

Hardware Description & Verification: SystemVerilog, Verilog, UVM, JasperGold, VCS, Verdi

Electronic Design Automation (EDA): Synopsys Design Compiler, PrimeTime, Cadence Innovus, Virtuoso, Vivado

Architectures & Simulators: RISC-V, ESP SoC Platform, ModelSim, Gem5

Programming & Scripting: C/C++, Python (NumPy, pandas, matplotlib), Java, C#, TCL, Bash

Grants & Funding

- **Undergraduate Research Opportunities Program (UROP) Grant** | *Univ. of Minnesota* May 2024
- Awarded \$2,500 to fund datacenter carbon modeling research. Co-Investigator: Senecy Zhang.
- **Student Research Fund** | *Grinnell College* March 2023
- Awarded \$4,000 to conduct independent academic research.